

ADCA — C Programming Test

Time: 1 Hour

Maximum Marks: 80

General Instructions

1. Section A is compulsory.
 2. Attempt **any 5 questions** from Section B.
 3. Attempt **any 5 questions** from Section C.
 4. Write programs with proper syntax and indentation.
 5. Assume suitable values wherever required.
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Section A — Multiple Choice Questions

30 × 1 = 30 Marks

Q1. Which of the following is the first phase of the Program Development Life Cycle?

- A) Coding
- B) Problem Definition
- C) Testing
- D) Maintenance

Q2. An algorithm is:

- A) A programming language
- B) A step-by-step procedure to solve a problem
- C) A type of compiler
- D) A flowchart symbol

Q3. Which flowchart symbol is generally used for a decision?

- A) Rectangle
- B) Oval
- C) Diamond
- D) Parallelogram

Q4. Which of the following is a low-level language?

- A) C
- B) Python
- C) Assembly Language
- D) Java

Q5. Which software translates a high-level language program into machine language?

- A) Editor
- B) Compiler
- C) Browser
- D) Operating System

Q6. Which of the following is a valid C identifier?

- A) 2value
- B) float
- C) total_marks
- D) total-marks

Q7. Which of the following is a C keyword?

- A) number
- B) main
- C) int
- D) marks

Q8. Which of the following represents a floating-point literal?

- A) 100
- B) 10.25
- C) "10.25"
- D) 'A'

Q9. Which data type is generally used to store a single character?

- A) int
- B) float
- C) char
- D) double

Q10. Which symbol is used to write a multi-line comment in C?

- A) `/**/`
- B) `//`
- C) `#`
- D) `--`

Q11. Which format specifier is used to print a float using `printf()`?

- A) %c
- B) %d
- C) %f
- D) %s

Q12. Which format specifier is used to read a character using scanf()?

- A) %c
- B) %d
- C) %f
- D) %s

Q13. Which operator gives the remainder after integer division?

- A) /
- B) %
- C) //
- D) &

Q14. Which of the following is a unary operator?

- A) +
- B) *
- C) ++
- D) &&

Q15. Which of the following is a logical AND operator in C?

- A) &
- B) &&
- C) ||
- D) !&

Q16. What will be the output?

```
int a = 5, b = 2;
```

```
printf("%d", a + b * 2);
```

- A) 14
- B) 9
- C) 12
- D) 7

Q17. What will be the output?

```
int x = 10;  
  
printf("%d", x++);
```

- A) 9
- B) 10
- C) 11
- D) Error

Q18. Which statement is used when a condition needs to be checked before executing a block?

- A) if
- B) switch
- C) goto
- D) continue

Q19. Which loop is guaranteed to execute its body at least once?

- A) for
- B) while
- C) do-while
- D) None

Q20. Which statement skips the remaining statements of the current loop iteration?

- A) break
- B) continue
- C) goto
- D) return

Q21. Which declaration represents a two-dimensional integer array?

- A) int a(2,3);
- B) int a[2,3];
- C) int a[2][3];
- D) array int a[2][3];

Q22. In C, a string is stored as:

- A) An integer
- B) A character array

- C) A floating-point array
- D) A structure only

Q23. Which function is used to find the length of a string?

- A) strcpy()
- B) strlen()
- C) strlen()
- D) length()

Q24. Which C feature allows us to group related variables of different data types under one name?

- A) Array
- B) Loop
- C) Structure
- D) Operator

Q25. Guess output

```
int a = 10, b = 3;  
printf("%d", a % b);
```

- A) 0
 - B) 1
 - C) 3
 - D) 3.33
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26. Prefix/Postfix

```
int a = 5;  
printf("%d ", a++);  
printf("%d", a);
```

- A) 5 5
- B) 5 6
- C) 6 6
- D) 6 5

27. if-else

```
int x = 10;  
if(x > 5)  
    printf("A");
```

else

```
    printf("B");
```

- A) A
- B) B
- C) AB
- D) Error

28. Loop

```
int i;
```

```
for(i = 1; i <= 3; i++)
```

```
{
```

```
    printf("%d ", i);
```

```
}
```

- A) 0 1 2
- B) 1 2 3
- C) 1 2
- D) 0 1 2 3

29. continue

```
int i;
```

```
for(i = 1; i <= 5; i++)
```

```
{
```

```
    if(i == 3)
```

```
        continue;
```

```
    printf("%d ", i);
```

```
}
```

- A) 1 2 3 4 5
- B) 1 2 4 5
- C) 1 2
- D) 3 4 5

30. Array

```
int a[4] = {10, 20, 30, 40};
```

```
printf("%d", a[2]);
```

- A) 10
- B) 20
- C) 30
- D) 40

Section B — One-Line / Very Short Answer

Attempt Any 5 — $5 \times 2 = 10$ Marks

- Q31.** What is an algorithm? Write one characteristic of a good algorithm.
- Q32.** Differentiate between a keyword and an identifier with one example each.
- Q33.** What is a variable? Give its general syntax for declaration.
- Q34.** Write the difference between printf() and scanf().
- Q35.** What is the difference between prefix and postfix increment operators?
- Q36.** What is the difference between while and do-while loops?
- Q37.** What is the purpose of the break and continue statements?
- Q38.** What is a 2D array? Give its declaration syntax.
- Q39.** What is a string in C? Why is \0 important?
- Q40.** What is a function? Write any two advantages of using functions.
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Section C — Programming Questions

Attempt Any 5 — $5 \times 8 = 40$ Marks

These are intentionally **basic to intermediate**.

Q41. Basic Input/Output + Operators

Write a C program to input two integers and display their:

- Sum
- Difference
- Product
- Quotient
- Remainder

Q42. Conditional Statement

Write a C program to input a student's marks and display:

- A for marks 80 or above
- B for marks 60–79
- C for marks 40–59
- Fail for marks below 40

Use an **else-if ladder**.

Q43. Nested Condition

Write a C program to input three numbers and find the **largest number** using if-else.

Q44. Loop

Write a C program to input a number and print its **multiplication table from 1 to 10** using a for loop.

Q45. Loop + Logic

Write a C program to input a positive integer and find the **sum of all numbers from 1 to that number** using a loop.

Example:

Input: 5

Output: 15

Q46. Switch Statement

Write a C program using switch to create a simple calculator.

The user should enter two numbers and an operator:

+, -, *, /

Display the appropriate result.

Q47. 1D Array

Write a C program to input **5 integers into a one-dimensional array** and find the **largest element**.

Q48. 2D Array / Matrix

Write a C program to input a **2 × 2 matrix** and display the matrix in proper row and column format.

Example:

1 2

Q49. String

Write a C program to input a string and find its **length using strlen()**. Also display the entered string.

Q50. Function

Write a C program using a **user-defined function** to calculate the square of a number.

For example:

Input: 5

Output: Square = 25